

L10 Monotone Allocations and Myerson's Lemma

CS 280 Algorithmic Game Theory

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Inspired and some figures by Tim Roughgarden notes

Recap

Three desirable **guarantees**

1. **DSIC**: Truthful bidding is a dominant strategy.

Easy to play for bidders, **Predict** outcome.

2. Social **surplus maximization**:

$$\sum_{i=1}^n x_i v_i$$

where x_i is the amount allocated to i .

3. The auction can be implemented in **polynomial time**.

An Example: Sponsored Search Auctions

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- Items for sale are **k “slots”**
- **Bidders** are the **advertisers**.
- Each slot j has CTR (click-through-rate) a_j .
- Each bidder i has private **valuation** v_i and gets value $a_j \cdot v_i$. Note $a_1 \geq \dots \geq a_k$

Probability
to get a click



Definitions

Definition (**Single parameter environments**). *A single parameter environment is defined:*

- *n bidders with private v_i ,*
- ***Feasible set** $\mathcal{X}$, each element of which is a n -dimensional vector $(x_1, \dots, x_n)$ in which x_i is the amount of "stuff" given to i .*

Examples:

1. **Single-item** auctions: $\mathcal{X}$ is 0-1 vectors with **at most one** 1, i.e., $\sum x_i \leq 1$.
2. **k identical goods**, each bidder gets **at most one**: $\mathcal{X}$ is 0-1 vectors with $\sum x_i \leq k$.
3. In sponsored search, $\mathcal{X}$ is the set of n -vectors with x_i being a_j if slot j is assigned to bidder i .

More Definitions

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Definition (Allocation and Payments). *A sealed-bid auction is defined:*

1. *Bidders report bids $b = (b_1, \dots, b_n)$,*
2. *Auctioneer chooses feasible allocation $x(b) \in \mathcal{X}$.*
3. *Auctioneer chooses payments $p(b) \in \mathbb{R}^n$.*
4. *Bidder i gets utility $u_i = v_i \cdot x_i(b) - p_i(b)$.*

Monotone Allocations and Myerson's Lemma

Definition (Monotone Allocations). *An allocation rule x for a single-parameter environment is **monotone** if for every bidder i and bids b_{-i} by rest of bidders, the allocation*

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Theorem (Myerson's Lemma). *Let (x, p) be a mechanism. We assume that $p_i(b) = 0$ whenever $b_i = 0$, for all bidders i .*

- 1. It holds that if (x, p) is DSIC mechanism then x is **monotone**.*
- 2. If x is a monotone allocation, then there is a unique payment rule such that (x, p) is DSIC.*

Myerson's Lemma: Monotone

Proof. Suppose (x, p) is a DSIC and let $0 \leq y \leq z$.

If bidder i has **private valuation** z , to avoid reporting y , DSIC demands

$$z \cdot x_i(z) - p_i(z) \geq z \cdot x_i(y) - p_i(y) \text{ for all } i.$$

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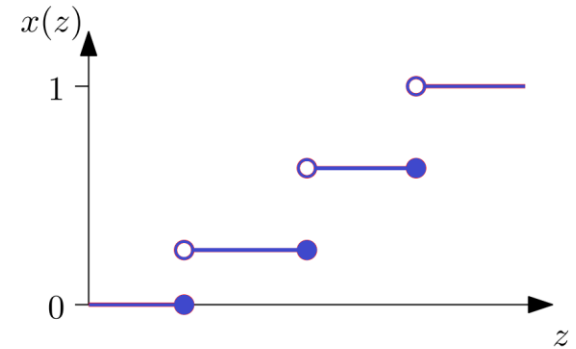
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Proof cont. Assume x is monotone for the rest of the proof and x is piecewise constant (**simple function**). if there is a jump at z (say of magnitude h) then as $y \rightarrow z$ from left we get

$$z \cdot h \leq p(y) - p(z) \leq y \cdot h.$$



Myerson's Lemma: Payments

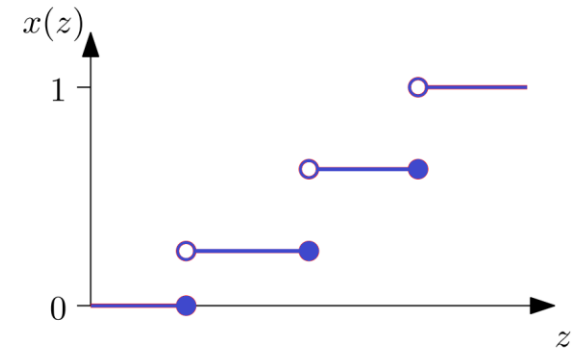
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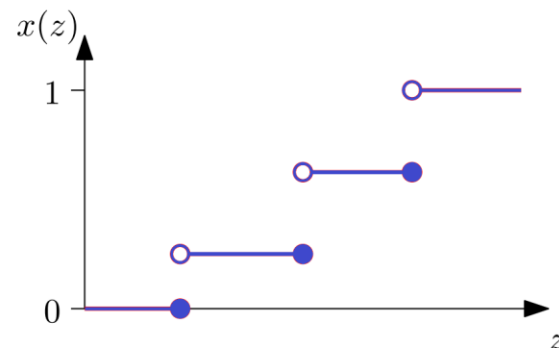
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We conclude that (given $p_i(0) = 0$)

$$p_i(b_i, b_{-i}) = \sum_{j=1}^l z_j \cdot \text{jump in } x_i(., b_{-i}) \text{ at } z_j,$$

where $z_1, \dots, z_l$ are the **breakpoints** of $x_i(., b_{-i})$ in $[0, b_i]$.



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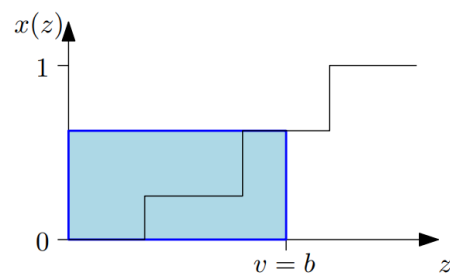
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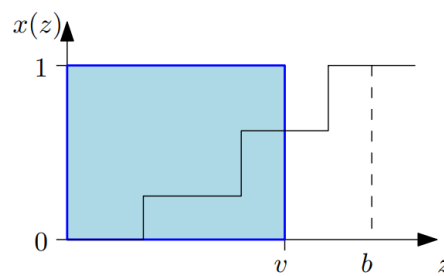
$$p_i(b_i, b_{-i}) = \int_0^{b_i} z \cdot \frac{dx_i(z, b_{-i})}{dz} dz.$$

Myerson's Lemma: DSIC

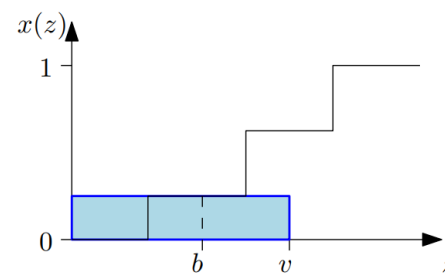
Proof cont. By picture.



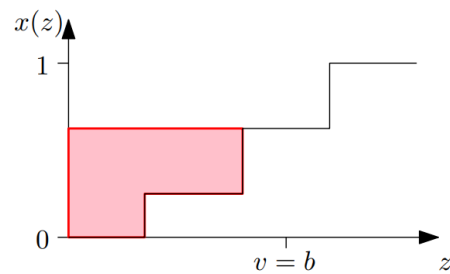
(a) $v \cdot x(v)$



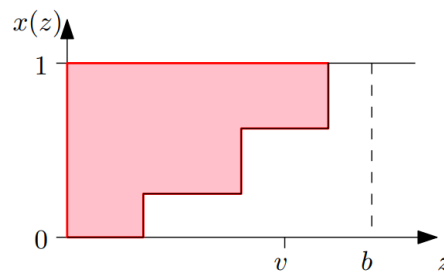
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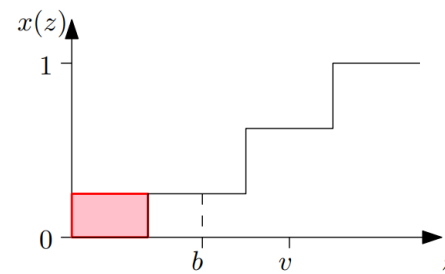
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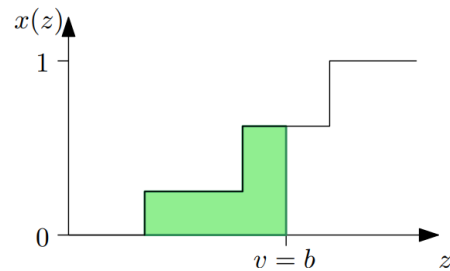
(d) $p(v)$



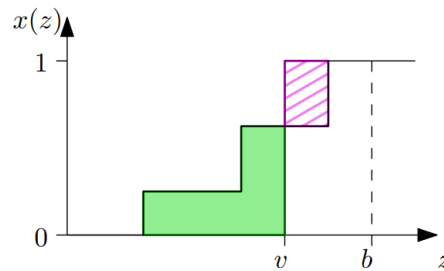
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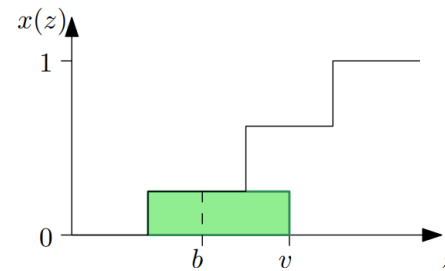
(f) $p(b)$ with $b < v$



(g) utility with $b = v$



(h) utility with $b > v$



(i) utility with $b < v$

Back to Sponsored Search Auctions

Remark: Myerson's Lemma regenerates the **Vickrey auction as a special case**. (why?)

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Answer: Fix i, b_{-i} and set $B = \max_{j \neq i} b_j$. Then $x_i(z, b_{-i})$ is 0 for $0 \leq z < B$ and 1 for $z \geq B$. Moreover, $p_i(z, b_{-i}) = B$ for $z \geq B$ and 0 for $0 \leq z < B$.

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- **Step 1: Assume**, without justification, that bidders **bid truthfully**. How should we assign bidders to slots so that we **can maximize surplus**?
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$$p_i(b) = \sum_{j=i}^k b_{j+1} (a_j - a_{j+1})$$